CloudWatch AI Control Plane

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Executive Summary

Enterprise AI adoption is outpacing governance: 72% of organizations run agentic AI in production, yet 60% lack governance. AWS customers building on Bedrock, SageMaker, and third-party models have no unified console to monitor AI health, enforce guardrails, track costs, or measure business impact. CloudWatch AI Control Plane fills this gap — a single console inside CloudWatch that unifies AI asset discovery, guardrail monitoring, cost optimization, and adoption maturity scoring. Unlike ServiceNow (on top of cloud), this is cloud-native, IAM-integrated, and cross-account by default.

1. Customer Problem

Target Persona: Maya Chen

Primary: Maya Chen, Sr. Cloud Operations Lead at a mid-size fintech (800 employees)

Maya manages 4 SREs responsible for all AWS infrastructure. Over 18 months, 6 engineering teams deployed AI features — fraud detection (SageMaker), support chatbot (Bedrock/Claude), code review agent (OpenAI API), and 3 internal tools using various LLMs. Maya knows about 4 of 6 deployments. She has no unified view of what AI is running, what it costs, whether guardrails are in place, or if any model is behaving unexpectedly. She spends ~5 hours/week manually correlating AI data across consoles.

Target Persona: Raj Patel

Secondary: Raj Patel, CISO at a healthcare SaaS company (2,000 employees)

Needs to ensure no AI model processes PHI without guardrails, prompt injection defenses are active, and the company can demonstrate AI governance to auditors. Today relies on quarterly manual reviews.

Jobs to Be Done

JTBD #1 (weekly, high exec visibility): When my VP asks 'what's our AI spend and what are we getting for it,' I want to pull up a single dashboard showing cost by team/model correlated with business metrics.

JTBD #2 (monthly, high compliance risk): When I discover a new AI deployment I didn't know about, I want to immediately see its guardrail status, cost footprint, and usage patterns.

JTBD #3 (daily, ops teams): When a Bedrock guardrail triggers, I want to be alerted immediately with context — which model, what content, how often.

JTBD #4 (quarterly, high dollar impact): When evaluating switching from GPT-4 to a cheaper model, I want cost vs. quality tradeoffs with real data.

JTBD #5 (semi-annual, extremely painful): When preparing for compliance audit, I want to generate a report showing all AI assets, governance status, and enforcement history.

Problem Depth

Root cause: AWS built AI services (Bedrock, SageMaker) and monitoring primitives (CloudWatch, CloudTrail) independently. No experience layer connects them into an AI-specific operations view.

Cost of status quo: ~5 hours/week manual correlation. 62% of CIOs compromising on governance (Logicalis 2026). 20-40% estimated AI overspend. 54% of COOs cite regulatory uncertainty.

2. Solution Proposal

AI Asset Discovery: Automatically discover all AI workloads across AWS Organization — Bedrock, SageMaker, Lambda functions calling AI APIs, ECS/EKS with AI SDK dependencies.

Unified AI Dashboard: Single-pane view of health (latency, errors, throughput), cost (tokens, compute, by team/model), and governance status (guardrail coverage, compliance).

Guardrail Monitoring: Real-time Bedrock Guardrails enforcement visibility — what's blocked, how often, which applications trigger guardrails.

Cost Intelligence: AI-specific cost breakdown with optimization recommendations (e.g., 'switch GPT-4 to Haiku, save $2,400/month with comparable accuracy').

Adoption Maturity Score: 5-level prescriptive model scoring governance posture with concrete next actions.

Scope

V1 In scope: Bedrock + SageMaker discovery, unified dashboard, guardrail monitoring/alerting, cost attribution, maturity assessment, cross-account via Organizations.

V1 Out of scope: Non-AWS AI discovery (v2), automated enforcement for non-Bedrock (v2), business outcome correlation (v2), kill switches (v2), model A/B testing (v3).

Competitive Differentiation

vs. ServiceNow: Cloud-native (IAM-integrated, no vendor creds), cross-account by default, zero-instrumentation for Bedrock, usage-based pricing vs. enterprise license. vs. Datadog: Governance built in (not just monitoring), discovery (not just instrumentation).

3. Success Metrics

Metric	Type	Target
% AI workloads with guardrail coverage	North Star	80% within 6 months
Time to discover new AI deployment	Supporting	< 24 hours (auto)
Time to generate compliance report	Supporting	< 5 minutes
AI cost visibility	Supporting	> 90% spend attributed
Console adoption	Supporting	500 MAU in 6 months
Alert false positive rate	Anti-metric	< 10%
Time to first value	Anti-metric	< 30 minutes

4. FAQs (25 MECE)

Category: Customer & Problem

Q1: Who is the primary buyer?

Cloud Operations or Platform Engineering lead at orgs with 3+ AI workloads. Secondary buyer: CISO/compliance officer. 76% of CIOs consider unchecked AI a serious concern (Logicalis 2026).

Q2: Why can't customers build custom CloudWatch dashboards?

They can for Bedrock, but custom dashboards require manual config per workload, provide no AI discovery, no guardrail enforcement patterns, no maturity assessment. 10 AI workloads across 5 accounts = manual dashboard hell.

Q3: What evidence shows customers want this?

72% have AI in production but 60% lack governance. 62% of CIOs compromise on governance due to lack of tools. ServiceNow invested in Traceloop + Veza acquisitions, validating the market.

Q4: Is the governance gap real?

Convergent data: Cisco 12% mature governance, Writer 79% face challenges, Agentic AI Institute 60% gap. Three independent surveys, same finding.

Category: Solution & Approach

Q5: Why build inside CloudWatch?

500K+ active customers already there. Zero new console, auto IAM integration, consolidated billing. ServiceNow bet standalone — we bet embedded.

Q6: How does AI asset discovery work?

Three sources: CloudTrail events (Bedrock/SageMaker API calls), Service Catalog tags, CloudWatch metric namespaces. Continuous scanning, new deployments surface in hours.

Q7: What does the maturity model look like?

5 levels from Ad Hoc to Leading, auto-assessed from observable signals. Each level has specific recommended actions. Prescriptive, not just diagnostic.

Category: Scope & Boundaries

Q8: Why is non-AWS AI discovery out of v1?

CloudTrail pattern matching for third-party APIs has high false-positive potential. V1 ships high-accuracy (Bedrock/SageMaker native). V2 adds third-party with confidence scores.

Q9: What's the phased plan?

V1: native discovery + dashboard + guardrails + cost + maturity. V2 (+3mo): third-party detection, custom policies, business outcomes. V3 (+6mo): kill switches, A/B testing, governance-as-code.

Category: Competitive & Market

Q10: How does ServiceNow AI Control Tower compare?

30+ integrations, 5 pillars, kill switches, Traceloop+Veza acquisitions. Most complete offering. Our edge: cloud-native, zero-instrumentation, cross-account default, usage pricing vs. enterprise license. They win multi-cloud; we win AWS-primary.

Q11: Is Datadog a threat?

Best developer-focused LLM observability. But no governance, no discovery, no compliance. Different buyer: Datadog serves developers, we serve ops and compliance. Can coexist.

Q12: What about Dynatrace?

Positioned as 'AI control plane' but expensive, complex, partnered with ServiceNow for enforcement. Our advantage: native AWS integration they can't match.

Category: Metrics & Success

Q13: Why guardrail coverage as North Star?

Directly measures the governance gap we exist to close. Actionable, measurable, correlated with customer value. Better than vanity (MAU) or lagging (cost savings) alternatives.

Q14: How to measure without cannibalization?

AI Control Plane is additive. Track MAU, workloads discovered, alerts configured, reports generated. Anti-metric: if Bedrock console traffic drops, that's consolidation, not cannibalization.

Category: Technical & Architecture

Q15: What AWS services does this integrate with?

Bedrock, SageMaker, CloudWatch, CloudTrail, Organizations, IAM, Cost Explorer, Resource Groups. All existing APIs — no new service dependencies.

Q16: How does cross-account work?

CloudWatch cross-account observability + Organization-level roles. Central admin sees all AI workloads. Guardrail policies pushed via CloudFormation StackSets. This is the AWS-native moat.

Q17: Data architecture?

No new data store. Reads from CloudWatch Metrics, CloudTrail, Bedrock Guardrails API, Cost Explorer in real-time. Only new storage: lightweight DynamoDB inventory cache per Organization.

Category: Business & Strategy

Q18: What's the TAM?

AI observability market $1.1B (2025), $3.29B by 2035. AWS-specific SAM: ~150K accounts using Bedrock/SageMaker at $500-2K/year = $75M-300M. Obtainable in 2 years at 5-10% adoption: $4M-30M. Strategic value exceeds direct revenue.

Q19: How does this affect pricing?

Included in CloudWatch usage pricing. Discovery and maturity scoring are free (drive deeper CloudWatch usage). Net positive: customers who optimize AI spend within AWS retain more total AWS spend.

Category: Risks & Mitigation

Q20: Discovery false positives?

V1 limited to native services (near 100% accuracy). V2 adds third-party with confidence scores and human confirmation. Manual 'add asset' fallback.

Q21: Maturity model perceived negatively?

Frame as progression, not grades. 'Next recommended action' not 'you're failing.' Anonymized peer benchmarks. Opt-in visible.

Q22: Biggest technical risk?

Cross-account aggregation latency at 100+ accounts. Mitigation: 15-minute cached refresh + real-time alerts via CloudWatch Alarms.

Q23: Yet another dashboard?

Not a dashboard you build — a console page you land on. Auto-configured, opinionated. Think 'EC2 instances page' not 'dashboard builder.'

Q24: Guardrails too narrow?

V1 acknowledges Bedrock-only limitation. V2 introduces Custom AI Policies for non-Bedrock via Lambda evaluation functions.

Q25: ServiceNow makes Control Tower free?

Their enterprise license model makes free unlikely. Our moat: no new vendor, no new credentials, no security review. Real risk is Datadog adding governance.

5. Risks & Open Questions

Risk	Likelihood	Impact	Mitigation
Discovery false positives	Medium	High	V1 native services only; confidence scores v2
Cross-account latency	Medium	Medium	15-min cached refresh + real-time alerts
Dashboard fatigue	Medium	Medium	Auto-configured page, zero setup
ServiceNow gains AWS integration	Low	High	Accelerate v1; deepen Orgs integration
Maturity model backlash	Medium	Low	Progression framing; peer benchmarks
Guardrails too narrow	High	Medium	Custom AI Policies in v2

Open Questions

1. Pricing model: included or incremental CloudWatch charges? — Owner: Product/Finance
2. Bedrock team alignment: complementary or competitive? — Owner: PM (cross-team)
3. Customer validation: need 5-10 interviews on AI governance pain — Owner: PM/UXR
4. SageMaker integration depth: Model Monitor in v1? — Owner: PM/Eng
5. EU AI Act: maturity model alignment needed? — Owner: Legal